Notes: Baugh, Ben-David, and Park (JF, 2018)

Can Taxes Shape an Industry? Evidence from the Implementation of the "Amazon Tax"

Contents

1 Big picture

- **Question.** The paper asks whether forcing Amazon to collect sales tax at checkout changes consumer purchasing behavior, Amazon revenues, and substitution to competing retailers.
- **Setting.** The empirical setting is the staggered implementation of state "Amazon Tax" laws from 2011 to 2015. These laws required Amazon to begin collecting sales tax from customers in adopting states.
- **Data.** The authors use transaction-level data from an online financial account aggregator. The broad data set covers 2.7 million households, and the main analysis focuses on 275,437 households that spent more than $200 on Amazon in 2011.
- **Main result.** After Amazon begins collecting sales tax, treated households reduce tax-exclusive Amazon spending by $3.65 per month, or 9.4% relative to pre-treatment mean spending. This implies an elasticity of roughly -1.2.
- **Tax-rate result.** A one-percentage-point increase in the local sales tax rate is associated with a $0.54 decline in monthly tax-exclusive Amazon spending, implying an elasticity of about -1.4.
- **Purchase-size result.** The response is much stronger for large purchases. Transactions of at least $250 fall by 29.1%, implying an elasticity of about -3.9. Elasticities become even larger for the highest transaction-size bins.
- **Household heterogeneity.** Lower-income households have larger percentage responses than higher-income households. Heavy Amazon shoppers have larger dollar responses, but percentage responses are similar across Amazon-shopping terciles.
- **Substitution.** The paper finds evidence of substitution to Newegg, an online electronics competitor, but not to Best Buy, eBay, PayPal, generic ".com" merchants, or Amazon Marketplace. Amazon's share of retail spending also falls.
- **Economic takeaway.** Online consumers are meaningfully tax-sensitive, especially for high-dollar purchases where avoiding tax saves more money. Sales-tax collection can reduce the price advantage of a dominant online retailer and redirect some spending to competitors.

2 Incremental contribution of the paper

- **From tax sensitivity to a policy treatment.** Prior work shows that online shoppers respond to sales-tax incentives. This paper studies a real, permanent policy change affecting Amazon, the largest U.S. online retailer.
- **Staggered state-level adoption.** The staggered rollout across 19 states creates a difference-in-differences setting in which treated households can be compared with households in states whose Amazon-tax status did not change.
- **Transaction-level household evidence.** Instead of using aggregate sales or survey responses, the paper observes household transactions across merchants and over time.
- **Tax-exclusive revenue measurement.** The authors adjust post-implementation transactions by local tax rates to measure Amazon spending net of sales tax. This directly targets the value of goods purchased from Amazon rather than the tax-inclusive amount paid by households.

- **Heterogeneity by transaction size.** A central contribution is showing that consumers are much more elastic for large purchases. This is important because the dollar value of tax avoidance grows with purchase size.
- **Competitor and substitution evidence.** The paper goes beyond Amazon sales and asks whether lost Amazon spending appears at competing retailers, especially in electronics where product comparison is relatively easy.
- **Policy relevance.** The results speak directly to state tax policy, online-commerce regulation, and the competitive advantage created by noncollection of sales tax.

3 Data and measurement

3.1 Sales-tax background

- Under the historical U.S. sales-tax nexus rules, out-of-state online retailers generally collected sales tax only in states where they had a substantial physical presence.
- Many states tried to broaden the definition of nexus through affiliate programs or subsidiaries, leading Amazon to begin collecting sales tax in some states.
- Before and during the sample, use-tax laws technically required consumers to self-report unpaid sales tax in many states, but compliance was very low.
- Between January 2011 and May 2015, 19 states implemented Amazon-tax collection during the authors' sample period.

Interpretation: The policy shock is not a change in the statutory tax rate itself. It is a change in tax collection salience and enforcement at checkout. Consumers who previously bought from Amazon without tax collection now saw the tax applied directly at purchase.

3.2 Household transaction data

- The raw data come from an online account aggregator that consolidates bank, debit-card, and credit-card transactions.
- The data include transaction dates, amounts, merchant descriptions, and household identifiers.
- The authors infer household residence from the state in which at least 75% of geographically identified transactions occur, then assign a city using the most common city in that state.
- The main sample includes households that spent more than $200 on Amazon during 2011, before any of the treatment states in the main rollout had implemented the Amazon Tax.

Interpretation: The transaction data are unusually rich for measuring online purchases, but the sample is not a fully random sample of U.S. households. It is likely younger and more urban than the general population.

3.3 Treatment timing and states

- Treated states include, among others, Texas, Pennsylvania, California, Arizona, New Jersey, Virginia, Georgia, Connecticut, Massachusetts, Wisconsin, Indiana, Nevada, Tennessee, North

Carolina, Florida, Maryland, Minnesota, and Illinois.

- States that had Amazon tax collection before the sample or did not change status during the sample serve as controls, depending on the specification.
- The event time for each treated household is the date when Amazon began collecting sales tax in that household's state.

Interpretation: The staggered timing is crucial. It allows the paper to separate a general time trend in Amazon spending from the state-specific treatment effect.

3.4 Outcome variables

- The main outcome is monthly tax-exclusive Amazon spending per household.
- For treated states after implementation, the authors divide Amazon transaction amounts by one plus the local sales-tax rate.
- Large-purchase outcomes restrict the monthly total to transactions at or above $250, or to transaction-size bins such as $0.01-$99.99, $100-$199.99, and so on.
- Substitution outcomes include spending at Best Buy, Newegg, eBay, PayPal, generic online merchants, Amazon Marketplace, and the Amazon share of retail spending.

Interpretation: Tax-exclusive spending is the correct revenue concept for Amazon. Tax-inclusive spending would mechanically include the tax collected on behalf of state governments.

4 Models and empirical specifications

4.1 Baseline difference-in-differences specification

The baseline household-month regression is:

$$Y_{h,t} = \beta_0+\beta_1\left(TreatedState_h \times I(t \geq Q)_{h,t}\right)+\beta_2 CostOfLivingIndex_{c,t}+\text{Month FE}_t+\text{Household FE}_h+\varepsilon_{h,t}.$$

- $Y_{h,t}$ is monthly tax-exclusive Amazon spending for household h in month t.
- $TreatedState_h \times I(t \geq Q)_{h,t}$ equals one for treated-state households after Amazon begins collecting tax in that state.
- Household fixed effects absorb time-invariant differences in households, including baseline Amazon preference.
- Year-month fixed effects absorb aggregate trends in online retail and macroeconomic conditions.
- The city-month cost-of-living index controls for local spending conditions using gas, restaurants, groceries, and non-Amazon retail purchases.

Interpretation: The coefficient β_1 measures the change in Amazon spending for treated households after tax collection begins, relative to the contemporaneous change for control households.

4.2 Timing specification

The paper also replaces the single post-treatment indicator with event-time indicators:

$$TreatedState_h \times I(t = Q - 1), \quad TreatedState_h \times I(t = Q0), \quad TreatedState_h \times I(t \geq Q + 1).$$

- The pre-quarter indicator tests whether households stock up before the tax becomes effective.
- The immediate post-quarter and later post-quarter indicators test whether the effect is short-lived or persistent.
- The event-study figure extends this logic to monthly event-time coefficients.

Interpretation: The observed pattern is a small pre-implementation buildup followed by a sharp and persistent decline in Amazon purchases.

4.3 Tax-rate interaction

A second elasticity specification interacts treatment with the local tax rate:

$$Y_{h,t} = \cdots + \beta_1 \left(TreatedState_h \times I(t \geq Q)_{h,t} \times TaxRate_h\right) + \cdots .$$

- This tests whether consumers in higher-tax jurisdictions reduce Amazon spending more.
- The estimated coefficient implies that a one-percentage-point tax increase lowers monthly Amazon spending by about \$0.54.
- Normalizing by mean spending yields an elasticity of about -1.4.

Interpretation: The tax-rate interaction strengthens the interpretation that the spending decline is a price/tax response, not just a general post-law behavioral change.

4.4 Heterogeneity specifications

- The authors split households into income terciles and into terciles based on 2011 Amazon spending.
- Low-income households have a larger percentage decline than high-income households.
- Heavy Amazon shoppers have the largest dollar declines, but the percentage declines are similar across Amazon-spending terciles.

Interpretation: The income result is consistent with lower-income households being more price-sensitive or more willing to search for alternatives.

4.5 Large purchases and transaction-size bins

- For transactions of at least \$250, the baseline treatment effect is much larger in percentage terms.
- The authors then estimate separate effects by purchase-size bins.
- Elasticities rise with transaction size, reaching about -6.8 for the \$700-and-above bin.

Interpretation: If search or switching costs are partly fixed, consumers will be most willing to avoid tax on large purchases because the dollar savings are larger.

4.6 Substitution regressions

- For each competing retailer, the paper uses the same treatment indicator and household-month structure.
- The key positive substitution result is for Newegg, whose spending increases by about 13% after the Amazon Tax.
- The Amazon share of retail purchases falls by 0.5 percentage points overall and by 0.7 percentage points among heavy Amazon shoppers.

Interpretation: The substitution evidence is strongest where products are comparable and the competing retailer preserves a tax advantage.

5 Identification and empirical design

5.1 What identifies the effect

- Identification comes from comparing treated households before and after Amazon begins collecting tax, relative to control households over the same calendar months.
- Household fixed effects control for persistent differences in households.
- Year-month fixed effects control for common aggregate shocks to Amazon or online retail.
- State-level staggered timing helps separate treatment effects from national online-shopping trends.

Interpretation: The design is strongest if treated and control households would have followed parallel trends in Amazon spending absent the tax-collection change.

5.2 Addressing nonrandom policy adoption

- States do not randomly adopt Amazon-tax laws. A concern is that weak state economies may both induce tax adoption and reduce Amazon spending.
- The paper checks state GDP growth around implementation and finds no significant post-implementation deterioration in treated states.
- It checks household income around implementation and finds no meaningful decline.
- It examines electronics-retailer spending and does not find a general consumption slowdown.
- The event-study pattern shows a sharp break in Amazon spending around implementation, supporting the policy interpretation.

Interpretation: These tests do not make the policy randomized, but they reduce the concern that the estimated effect is driven by local economic weakness.

5.3 Measurement and external validity

- The account-aggregator sample is likely younger and more urban than the U.S. population.

- The data identify merchants, but not specific products. This makes substitution harder to measure, especially for large retailers such as Walmart or Costco.
- The authors rely on inferred household location and local tax rates; robustness checks use alternative residence assignment methods.
- Use-tax compliance was very low, so the change in checkout collection likely represents a salient and enforceable price change for most consumers.

Interpretation: The paper has strong internal evidence for account-aggregator households, but external validity to all U.S. households should be read with caution.

6 Tables and figures

6.1 Figure 1 and Table I: tax importance and sample geography

Read:

- Figure 1 shows that general sales tax is an important revenue source for many states, with substantial variation across states.
- Table I compares the geographic distribution of the paper's sample with the U.S. Census. California and New York are overrepresented, while some Midwestern and smaller states are underrepresented.
- The table motivates robustness checks to make sure results are not driven by a few large states such as California.

Economic message: Sales-tax collection is fiscally important for states, and the sample is geographically broad enough for a state-level policy design, though not perfectly representative.

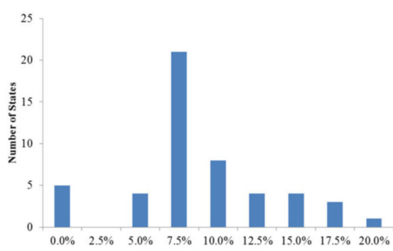

Figure 1. Histogram of (Sales Tax Revenue/Total State Revenue) for the 50 states in 2011. This figure illustrates the importance of sales tax revenues as a percentage of total state revenues. The data come from the 2011 U.S. Census Annual Survey of State and Local Government Finance: http://www.census.gov/govs/local/. This figure shows that the importance of this tax varies considerably across states, ranging from 0% of state revenues in states without a sales tax (such as Oregon and Alaska) to as high as 21.0% of state revenues for Washington. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 1: sales-tax revenue as a share of total state revenue

Table I
Geographic Distribution of the Sample

This table shows the geographic distribution of the households in the sample relative to the 2010 U.S. Census.

	% Households Residing				% Households Residing		
State	Data	U.S. Census	Difference	State	Data	U.S. Census	Difference
Alabama	0.6%	1.5%	−1.0%	Montana	0.1%	0.3%	−0.2%
Alaska	0.3%	0.2%	0.0%	Nebraska	0.3%	0.6%	−0.3%
Arizona	1.8%	2.1%	−0.2%	Nevada	0.9%	0.9%	0.0%
Arkansas	0.3%	0.9%	−0.6%	New Hampshire	0.2%	0.4%	−0.2%
California	21.5%	12.1%	9.5%	New Jersey	2.1%	2.8%	−0.8%
Colorado	1.1%	1.6%	−0.5%	New Mexico	0.4%	0.7%	−0.2%
Connecticut	1.2%	1.2%	0.1%	New York	19.2%	6.3%	13.0%
Delaware	0.1%	0.3%	−0.1%	North Carolina	2.5%	3.1%	−0.6%
District of Columbia	0.4%	0.2%	0.2%	North Dakota	0.1%	0.2%	−0.1%
Florida	6.2%	6.1%	0.1%	Ohio	0.7%	3.7%	−3.0%
Georgia	2.6%	3.1%	−0.5%	Oklahoma	0.6%	1.2%	−0.6%
Hawaii	0.4%	0.4%	−0.1%	Oregon	0.7%	1.2%	−0.5%
Idaho	0.2%	0.5%	−0.3%	Pennsylvania	1.2%	4.1%	−2.9%
Illinois	5.4%	4.2%	1.3%	Rhode Island	0.2%	0.3%	−0.2%
Indiana	0.4%	2.1%	−1.7%	South Carolina	0.9%	1.5%	−0.6%
Iowa	0.2%	1.0%	−0.8%	South Dakota	0.1%	0.3%	−0.2%
Kansas	0.4%	0.9%	−0.5%	Tennessee	1.0%	2.1%	−1.0%
Kentucky	0.3%	1.4%	−1.1%	Texas	10.9%	8.1%	2.8%
Louisiana	0.4%	1.5%	−1.0%	Utah	0.3%	0.9%	−0.6%
Maine	0.2%	0.4%	−0.3%	Vermont	0.1%	0.2%	−0.1%
Maryland	2.4%	1.9%	0.5%	Virginia	4.1%	2.6%	1.5%
Massachusetts	2.8%	2.1%	0.6%	Washington	1.7%	2.2%	−0.4%
Michigan	0.7%	3.2%	−2.5%	West Virginia	0.1%	0.6%	−0.5%
Minnesota	0.4%	1.7%	−1.3%	Wisconsin	0.3%	1.8%	−1.5%
Mississippi	0.2%	1.0%	−0.8%	Wyoming	0.1%	0.2%	−0.1%
Missouri	0.8%	1.9%	−1.1%				

(b) Table I: geographic distribution of households

6.2 Figure 2 and Table II: sample income and raw before-after spending

Read:

- Figure 2 compares the income distribution in the transaction sample with Census income data. The sample broadly covers the income distribution but differs at the lowest-income and top-coded high-income ends.
- Table II reports raw average Amazon spending in the three months before and after implementation in treated and control states.
- Treated states generally show weaker post-implementation Amazon spending relative to controls, motivating the formal difference-in-differences regressions.

Economic message: The raw data already suggest a decline in treated-state Amazon spending, but the regression framework is needed to control for household, time, and cost-of-living differences.

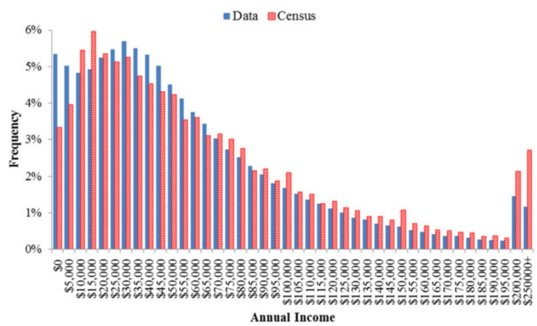

Figure 2. Distribution of annual income. This figure compares the distribution of annual income of our sample and the U.S. Census. The income observed in our data is that which arrives in households' checking and savings accounts. It therefore equals gross income minus the sum of withholdings (payroll tax, state tax, federal tax, healthcare contributions, retirement contributions, etc.). These omissions result in gross income that is higher than what we directly observe. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 2: annual income distribution

Table II

Average Monthly Tax-Exclusive Expenditures before and after Sales Tax Change

This summary table presents average tax-exclusive spending at Amazon in the ±3-month window before and after implementation of Amazon Tax laws. We include only those households that spent over $200 on Amazon during 2011. If an Amazon transaction occurs after the tax law changes and the household resides in one of the 19 affected states, we adjust the postimplementation transactions by dividing by one plus the local sales tax rate to create the tax-exclusive amount. Control states are the 31 states that do not change their Amazon Tax status during our sample period.

	States (3-Month Window)									
	All	TX	PA	CA	AZ	NJ	VA	GA	WV	CT
Before implementation										
Treated state(s)	$40.51	$32.45	$37.56	$37.21	$51.00	$36.31	$44.05	$36.75	$38.30	$42.75
Control states	$35.71	$30.72	$31.09	$31.19	$46.83	$33.66	$34.38	$34.38	$34.45	$34.66
After implementation										
Treated state(s)	$39.93	$29.98	$37.64	$44.06	$31.90	$35.10	$45.63	$37.58	$58.65	$59.82
Control states	$39.68	$31.32	$35.27	$45.52	$32.06	$34.45	$37.74	$37.74	$51.13	$51.89
	States (3-Month Window)									
	MA	WI	IN	NV	TN	NC	FL	MD	MN	IL
Before implementation										
Treated state	$41.14	$44.75	$60.87	$54.06	$61.45	$58.11	$38.91	$42.83	$44.97	$49.18
Control states	$34.66	$34.66	$51.13	$51.13	$51.13	$51.88	$35.23	$36.68	$36.68	$46.66
After implementation										
Treated state	$56.07	$60.42	$39.43	$33.69	$35.95	$35.65	$36.02	$54.73	$52.59	$31.42
Control states	$51.89	$51.89	$35.50	$35.50	$35.50	$35.23	$36.98	$47.85	$47.89	$33.41

(b) Table II: average monthly tax-exclusive expenditures before and after tax change

6.3 Tables III and IV: validity checks and the main Amazon-spending effect

Read:

- Table III checks whether treated states experience different GDP growth or household-income changes after tax implementation. The estimates are not consistent with a treated-state economic slowdown.
- Table IV is the core result: Amazon spending falls by $3.648 per month after implementation, equal to 9.4% of the treated pre-event mean.
- Table IV also shows a small pre-period buildup and a persistent post-period decline, plus a tax-rate interaction implying elasticity around -1.4.

Economic message: The main result is not easily explained by local income shocks. Households cut Amazon purchases when tax collection becomes salient and enforceable.

Table III
State GDP Growth and Household Income around Amazon Tax Implementation

This table explores whether states that implemented the Amazon Tax experienced different GDP growth (columns (1) and (2)) or a change in household income (columns (3) and (4)) than states that did not implement the tax. All regressions are ordinary least squares (OLS) regressions and include time and state fixed effects. The unit of observation in columns (1) and (2) is the state-quarter. The regression in column (1) is weighted by the GDP of each state. The regression in column (2) is weighted by the relative number of households in each state in the sample. The unit of observation in columns (3) and (4) is the household-month. Column (3) looks at household income after the implementation of the tax in the treated states. Column (4) looks at the short- and long-term changes in household income after the implementation of the tax in the treated states. Standard errors are clustered by state and time. *Treated State* is an indicator variable that takes the value of 1 for states that implemented an Amazon Tax during our sample period. $I(t \geq Q)$ is an indicator variable that takes the value of 1 for all months after implementation of the Amazon Tax. $I(t = Q_{-1})$, $I(t = Q_0)$, and $I(t \geq Q_{+1})$ are indicator variables for the quarter(s) before, the quarter after, and the subsequent quarters following tax implementation, respectively. *t*-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable	State-Level GDP Growth (%)		Income	
	(1)	(2)	(3)	(4)
Treated state × $I(t \geq Q)$	0.184	−0.104	58.224	
	(0.42)	(−0.22)	(1.68)	
Treated state × $I(t = Q_{-1})$				−3.130
				(−0.09)
Treated state × $I(t = Q_0)$				36.061
				(1.11)
Treated state × $I(t \geq Q_{+1})$				65.934
				(1.41)
State fixed effect	Yes	Yes	Yes	Yes
Time fixed effect	Yes	Yes	Yes	Yes
Weighting	GDP	#Households		
Observations	757	757	10,436,160	10,436,160
R^2	48%	52%	73%	73%

(a) Table III: state GDP growth and household income around implementation

Table IV
Effect of Amazon Tax on Tax-Exclusive Monthly Amazon Expenditures

This table explores the effect of the Amazon Tax on tax-exclusive Amazon expenditures. The unit of observation is the household-month, and the dependent variable is the sum of monthly Amazon transactions per household. *Treated State* is an indicator variable that takes the value of 1 for states that implemented an Amazon Tax during our sample period. $I(t \geq Q)$ is an indicator variable that takes the value of 1 for all months after implementation of an Amazon Tax. $I(t = Q_{-1})$, $I(t = Q_0)$, and $I(t \geq Q_{+1})$ are indicator variables for the quarter(s) before, the quarter after, and the subsequent quarters following the tax implementation, respectively. *Tax Rate* is the household's sales tax rate. To account for regional differences in cost of living that vary over time, we introduce a time-varying cost of living index at the city-month level, denoted *Cost of Living Index*. This index is computed by calculating the mean expenditures in the categories of gas, restaurants, groceries, and retail (excluding Amazon purchases) for each city-month. All regressions are ordinary least squares (OLS) regressions and include household and year-month fixed effects. Standard errors are clustered by state and time. *t*-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable	Amazon Spending (Tax-Exclusive)		
	(1)	(2)	(3)
Treated state × $I(t \geq Q)$	-3.648^{***}		
	(−5.07)		
Treated state × $I(t = Q_{-1})$		1.421^{***}	
		(2.87)	
Treated state × $I(t = Q_0)$		-3.289^{***}	
		(−3.82)	
Treated state × $I(t \geq Q_{+1})$		-3.208^{***}	
		(−4.47)	
Treated state × $I(t \geq Q)$ × Tax rate			-54.328^{***}
			(−7.05)
Household fixed effect	Yes	Yes	Yes
Year-month fixed effect	Yes	Yes	Yes
Cost of living index (city-month)	Yes	Yes	Yes
Observations	10,436,160	10,436,160	10,436,160
R^2	28%	28%	28%
Mean spending of treated	$39.00	$39.00	$39.00
Mean tax rate of treated	7.5%	7.5%	7.5%
Implied elasticity (Treated state × $I(t \geq Q)$)	−1.24		
Implied elasticity (Treated state × $I(t = Q_{-1})$)		0.48	
Implied elasticity (Treated state × $I(t = Q_0)$)		−1.12	
Implied elasticity (Treated state × $I(t = Q_{+1})$)		−1.09	
Implied elasticity (Treated state × $I(t \geq Q)$ × Tax rate)			−1.39

(b) Table IV: effect of Amazon Tax on tax-exclusive Amazon expenditures

6.4 Tables V and VI: household heterogeneity and large purchases

Read:

- Table V splits households by income and by historical Amazon spending. Low-income households show the largest percentage response, while high Amazon spenders show the largest dollar response.
- Table VI restricts the outcome to Amazon purchases of at least $250. The decline is $2.249 per month, equal to 29.1% of mean spending in that category.
- The implied elasticity for large purchases is about -3.9, far larger than the baseline elasticity for all purchases.

Economic message: Taxes matter most when the dollar tax bill is large. This supports a search-cost or switching-cost interpretation: consumers are most willing to look for alternatives when the potential tax saving is meaningful.

Table V

Effect of the Amazon Tax on Different Types of Households

This table explores the effect of the Amazon Tax on different types of households. The unit of observation is the household-month, and the dependent variable is the tax-exclusive sum of monthly Amazon transactions per household. Households are divided into three groups depending on their monthly income and total Amazon spending in 2011. *Treated State* is an indicator variable that takes the value of 1 for states that implemented an Amazon Tax during our sample period. $I(t \geq Q)$ is an indicator variable that takes the value of 1 for all months after implementation of the Amazon Tax. To account for regional differences in cost of living that vary over time, we introduce a time-varying cost of living index at the city-month level, denoted *Cost of Living Index*. This index is computed by calculating the mean expenditures in the categories of gas, restaurants, groceries, and retail (excluding Amazon purchases) for each city-month. All regressions are ordinary least squares (OLS) regressions and include household and year-month fixed effects. Standard errors are clustered by state and time. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable	Amazon Spending (Tax-Exclusive)					
	Income Terciles			Amazon Spending Terciles		
	High (1)	Mid (2)	Low (3)	High (4)	Mid (5)	Low (6)
Treated state × $I(t \geq Q)$	-3.755^{***}	-3.675^{***}	-3.038^{***}	-6.224^{***}	-2.830^{***}	-1.649^{***}
	(−3.53)	(−4.85)	(−5.81)	(−5.37)	(−4.82)	(−2.89)
Household fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Cost of living index (city-month)	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,501,723	2,501,759	2,501,788	3,478,700	3,478,723	3,478,737
R^2	30%	26%	24%	30%	20%	17%
Mean spending of treated	\$53.34	\$38.72	\$30.57	\$65.97	\$29.86	\$20.53
Mean tax rate of treated	7.4%	7.5%	7.6%	7.5%	7.5%	7.5%
Implied elasticity	−0.95	−1.27	−1.31	−1.25	−1.26	−1.07
Treated state × $I(t \geq Q)$/Mean spending	−7.0%	−9.5%	−9.9%	−9.4%	−9.5%	−8.0%

(a) Table V: effects by income and Amazon-spending terciles

Table VI

Effect of the Amazon Tax on Large Amazon Expenditures

This table explores the effect of the Amazon Tax on tax-exclusive large Amazon expenditures. The unit of observation is the household-month, and the dependent variable is the sum of monthly Amazon transactions per household that are at least \$250. *Treated State* is an indicator variable that takes the value of 1 for states that implemented an Amazon Tax during our sample period. $I(t \geq Q)$ is an indicator variable that takes the value of 1 for all months after implementation of the Amazon Tax. $I(t = Q_{-1})$, $I(t = Q_0)$, and $I(t \geq Q_{+1})$ are indicator variables for the quarter(s) before, the quarter after, and the subsequent quarters following the tax implementation, respectively. *Tax Rate* is the household's sales tax rate. To account for regional differences in cost of living that vary over time, we introduce a time-varying cost of living index at the city-month level, denoted *Cost of Living Index*. This index is computed by calculating the mean expenditures in the categories of gas, restaurants, groceries, and retail (excluding Amazon purchases) for each city-month. All regressions are ordinary least squares (OLS) regressions and include household and year-month fixed effects. Standard errors are clustered by state and time. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable	Amazon Spending ≥ \$250 (Tax-Exclusive)		
	(1)	(2)	(3)
Treated state × $I(t \geq Q)$	-2.249^{***}		
	(−7.92)		
Treated state × $I(t = Q_{-1})$		0.471^{**}	
		(2.16)	
Treated state × $I(t = Q_0)$		-2.128^{***}	
		(−6.78)	
Treated state × $I(t \geq Q_{+1})$		-2.103^{***}	
		(−7.04)	
Treated state × $I(t \geq Q)$ × Tax rate			-31.923^{***}
			(−11.31)
Household fixed effect	Yes	Yes	Yes
Year-month fixed effect	Yes	Yes	Yes
Cost of living index (city-month)	Yes	Yes	Yes
Observations	10,436,160	10,436,160	10,436,160
R^2	8%	8%	8%
Mean spending of treated	\$7.73	\$7.73	\$7.73
Mean tax rate of treated	7.5%	7.5%	7.5%
Implied elasticity (Treated state × $I(t \geq Q)$)	−3.87		
Implied elasticity (Treated state × $I(t = Q_{-1})$)		0.81	
Implied elasticity (Treated state × $I(t = Q_0)$)		−3.66	
Implied elasticity (Treated state × $I(t = Q_{+1})$)		−3.61	
Implied elasticity (Treated state × $I(t \geq Q)$ × Tax rate)			−4.13

(b) Table VI: effect on large Amazon purchases

6.5 Table VII and Figure 3: elasticities by transaction size

Read:

- Table VII estimates the treatment effect separately for transaction-size bins.
- Figure 3 plots the implied elasticities by purchase size.
- Elasticities become more negative as transaction size rises, with the largest bin showing an elasticity around -6.8.

Economic message: The result is not just that consumers dislike taxes. Consumers respond to the dollar amount of tax owed. Large-ticket purchases are where Amazon's pre-tax-collection advantage was most valuable.

Table VII

Elasticities as a Function of Purchase Size

This table explores how elasticity varies with purchase size. The dependent variable in column (1) is the tax-exclusive sum of monthly Amazon transactions per household. The dependent variable in columns (1) through (8) is the sum of tax-exclusive monthly Amazon transactions for various sized bins. Column (1) corresponds to purchases with tax-exclusive prices of $0.01 to $99.99, column (2) corresponds to purchases with tax-exclusive prices of $100.00 to $199.99, and so on. *Treated State* is an indicator variable that takes the value of 1 for states that implemented an Amazon Tax during our sample period. $I(t \geq Q)$ is an indicator variable that takes the value of 1 for all months after implementation of the Amazon Tax. To account for regional differences in cost of living that vary over time, we introduce a time-varying cost of living index at the city-month level, denoted *Cost of Living Index*. This index is computed by calculating the mean expenditures in the categories of gas, restaurants, groceries, and retail (excluding Amazon purchases) for each city-month. All regressions are ordinary least squares (OLS) regressions and include household and year-month fixed effects. Standard errors are clustered by state and time. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable	Tax-Exclusive Amazon Purchase Size in Bracket . . .							
	$0.01 to $99.99 (1)	$100.00 to $199.99 (2)	$200.00 to $299.99 (3)	$300.00 to $399.99 (4)	$400.00 to $499.99 (5)	$500.00 to $599.99 (6)	$600.00 to $699.99 (7)	$700 and Above (8)
Treated state × $I(t \geq Q)$	-1.245^{***}	-0.832^{***}	-0.533^{***}	-0.381^{***}	-0.183^{***}	-0.172^{***}	-0.157^{***}	-1.269^{***}
	(−3.03)	(−6.13)	(−7.73)	(−6.66)	(−4.70)	(−3.44)	(−7.08)	(−7.07)
Household fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cost of living index (city-month)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,436,160	10,436,160	10,436,160	10,436,160	10,436,160	10,436,160	10,436,160	10,436,160
R^2	38%	11%	6%	5%	5%	4%	3%	6%
Mean spending of treated	$24.22	$7.46	$3.30	$1.76	$1.10	$0.70	$0.47	$2.48
Mean tax rate of treated	7.5%	7.5%	7.5%	7.5%	7.5%	7.5%	7.5%	7.5%
Implied elasticity	−0.68	−1.48	−2.15	−2.88	−2.22	−3.28	−4.43	−6.79
Treated state × $I(t \geq Q)$/Mean spending	−5.1%	−11.2%	−16.2%	−21.7%	−16.7%	−24.7%	−33.3%	−51.1%

(a) Table VII: elasticities as a function of purchase size

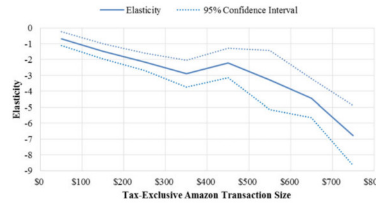

Figure 3. Elasticities as a function of purchase size. This figure presents the elasticities of Amazon shoppers as a function of tax-exclusive purchase size. The elasticities are imputed from regressions of tax-exclusive Amazon spending on an interaction term, which takes the value of 1 for treated states after the Amazon Tax begins. The dependent variable equals the purchase amount if it falls within the size bracket being investigated (e.g., $200 to $299.99), and 0 otherwise. The dashed lines indicate the 95% confidence interval, and standard errors are clustered by state and time. The last bucket ($700+) includes all of the tax-exclusive transactions that are greater than $700. To account for regional differences in cost of living that vary over time, we introduce a time-varying cost of living index at the city-month level. This index is computed by calculating the mean expenditures in the categories of gas, restaurants, groceries, and retail (excluding Amazon purchases) for each city-month. (Color figure can be viewed at wileyonlinelibrary.com)

(b) Figure 3: elasticity by purchase-size bin

6.6 Figure 4 and Table VIII: event-time dynamics and large-purchase heterogeneity

Read:

- Figure 4 shows monthly event-time coefficients before and after Amazon begins collecting tax. Both all Amazon spending and large-purchase spending drop sharply around implementation.
- The decline is persistent rather than a one-month adjustment.
- Table VIII shows that the large-purchase response is especially strong for low-income households, with a 34.3% reduction and implied elasticity of about -4.5.

Economic message: The event-study timing strengthens the causal interpretation. The large-purchase effect is not a gradual trend; it appears at the tax-collection event and remains negative.

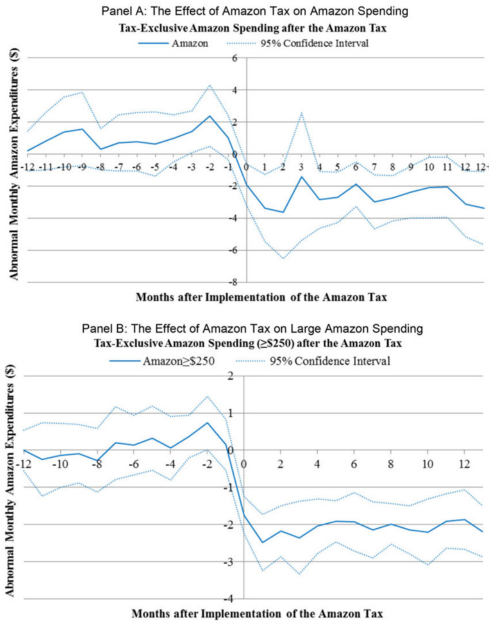

Figure 4. Amazon spending before and after the Amazon Tax. This figure illustrates the trend of the regression coefficients of monthly Amazon spending in the −12 to end-of-sample window surrounding implementation of Amazon Tax laws. The specification is similar to the base specification described previously but with a series of months-after-treatment indicator variables rather than quarters-after-treatment indicators. We run two different regressions. In Panel A, the dependent variable for the first regression is the sum of all tax-exclusive Amazon purchases. In Panel B, the dependent variable for the second regression is the sum of all tax-exclusive Amazon purchases that are at least $250. Regression coefficients for the two regressions are plotted. The dashed lines indicate the 95% confidence interval, and standard errors are clustered by state and time. To account for regional differences in cost of living that vary over time, we introduce a time-varying cost of living index at the city-month level. This index is computed by calculating the mean expenditures in the categories of gas, restaurants, groceries, and retail (excluding Amazon purchases) for each city-month. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 4: Amazon spending before and after tax implementation

Table VIII
Effect of the Amazon Tax on Different Types of Households for Large Purchases

This table explores the effect of the Amazon Tax on different types of households for large purchases. The unit of observation is the household-month, and the dependent variable is the tax-exclusive sum of monthly Amazon transactions per household that are at least $250. Households are divided into three groups depending on their monthly income and total Amazon spending in 2011. *Treated State* is an indicator variable that takes the value of 1 for states that implemented an Amazon Tax during our sample period. $I(t \geq Q)$ is an indicator variable that takes the value of 1 for all months after implementation of the Amazon Tax. To account for regional differences in cost of living that vary over time, we introduce a time-varying cost of living index at the city-month level, denoted *Cost of Living Index*. This index is computed by calculating the mean expenditures in the categories of gas, restaurants, groceries, and retail (excluding Amazon purchases) for each city-month. All regressions are ordinary least squares (OLS) regressions and include household and year-month fixed effects. Standard errors are clustered by state and time. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable	Amazon Spending ≥ $250 (Tax-Exclusive)					
	Income Terciles			Amazon Spending Terciles		
	High (1)	Mid (2)	Low (3)	High (4)	Mid (5)	Low (6)
Treated state × $I(t \geq Q)$	−2.601*** (−5.56)	−2.157*** (−6.12)	−1.879*** (−9.45)	−4.250*** (−7.59)	−1.467*** (−7.53)	−0.935*** (−4.51)
Household fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Cost of living index (city-month)	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,501,723	2,501,759	2,501,788	3,478,700	3,478,723	3,478,737
R^2	7%	6%	6%	9%	5%	5%
Mean spending of treated	$10.50	$6.83	$5.48	$14.12	$5.33	$3.59
Mean tax rate of treated	7.4%	7.5%	7.6%	7.5%	7.5%	7.5%
Implied elasticity	−3.34	−4.21	−4.53	−3.99	−3.66	−3.45
Treated state × $I(t \geq Q)$/ Mean spending	−24.8%	−31.6%	−34.3%	−30.1%	−27.5%	−26.0%

(b) Table VIII: large-purchase effects by household type

6.7 Table IX: substitution to competitors and Amazon retail share

Read:

- Panel A estimates spending changes at competing retailers. The clearest positive effect is for Newegg, where spending rises by $0.247 per month, or 13.0% of treated mean spending.
- Best Buy brick-and-mortar and online spending do not increase significantly, and neither do eBay, PayPal, or generic online merchants.
- Panel B shows that Amazon's share of retail spending falls by 0.5 percentage points overall, with a larger 0.7 percentage-point decline among heavy Amazon shoppers.

Economic message: Some lost Amazon sales appear to shift to tax-advantaged online substitutes, especially in electronics. But substitution is incomplete and difficult to detect when competitors sell many non-Amazon-like products.

Table IX

Substitution Effects from the Amazon Tax

This table explores the effect of the Amazon Tax on other retailers. Panel A investigates the dollar value spent at Best Buy, Newegg, eBay, PayPal, generic online merchants, and Amazon Marketplace. Panel B investigates the percentage of retail spending occurring at Amazon. In both panels, the unit of observation is the household-month. In Panel A, the dependent variable is the tax-inclusive sum of monthly retail transactions for a given retailer. Best Buy sales are categorized as either brick-and-mortar or online transactions. *DotCom* corresponds to a generic query intended to capture all other online merchants using the term ".com" in the description that are not otherwise classified in the other columns. We include households that spent at least $200 on Amazon during 2011. *Treated State* is an indicator variable that takes the value of 1 for states that implemented an Amazon Tax during our sample period. $I(t \geq Q)$ is an indicator variable that takes the value of 1 for all months after implementation of the Amazon Tax. To account for regional differences in cost of living that vary over time, we introduce a time-varying cost of living index at the city-month level, denoted *Cost of Living Index*. This index is computed by calculating the mean expenditures in the categories of gas, restaurants, groceries, and retail (excluding Amazon purchases) for each city-month. All regressions are ordinary least squares (OLS) regressions and include household and year-month fixed effects. Standard errors are clustered by state and time. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Dollar Value Substitution to Other Retailers

Dependent Variable	Best Buy (Brick) (1)	Best Buy (Online) (2)	Newegg (3)	eBay (4)	PayPal (5)	DotCom (6)	Amazon Marketplace (7)
Treated state × $I(t \geq Q)$	−0.018	−0.066	0.247***	0.030	−1.698	−0.271	−0.948*
	(−0.07)	(−0.53)	(2.99)	(1.22)	(−1.15)	(−0.19)	(−1.77)
Household fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cost of living index (city-month)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,436,160	10,436,160	10,436,160	10,436,160	10,436,160	10,436,160	10,436,160
R^2	8%	5%	12%	27%	26%	21%	27%
Mean spending of treated	$11.63	$2.28	$1.89	$0.51	$36.31	$58.89	$41.51
Mean tax rate of treated	7.5%	7.5%	7.5%	7.5%	7.5%	7.5%	7.5%
Treated state × $I(t \geq Q)$/Mean spending	−0.2%	−2.9%	13.0%	5.9%	−4.7%	−0.5%	−2.3%

(Continued)

(a) Table IX, Panel A: dollar substitution to other retailers

Table IX—*Continued*

Panel B: Ratio of Retail Spending at Amazon

Dependent Variable	Amazon/(Amazon + Other Retail)		
		Amazon Spending Terciles	
	Overall (1)	High (2)	Mid (3)
Treated state × $I(t \geq Q)$	−0.005***	−0.007***	−0.004***
	(−4.68)	(−4.48)	(−4.69)
Household fixed effect	Yes	Yes	Yes
Year-month fixed effect	Yes	Yes	Yes
Cost of living index (city-month)	Yes	Yes	Yes
Observations	9,592,627	3,214,499	3,197,020
R^2	28%	32%	24%

(b) Table IX, Panel B: Amazon share of retail spending

7 Short summary

- The paper studies the staggered adoption of Amazon sales-tax collection across U.S. states.
- The main sample uses household transaction data for 275,437 Amazon customers.
- Amazon spending falls by 9.4% after tax collection begins, implying elasticity around -1.2.
- A one-percentage-point higher tax rate lowers monthly Amazon spending by about $0.54, implying elasticity around -1.4.
- Large purchases are far more sensitive: purchases of at least $250 fall by 29.1%, with elasticity around -3.9.
- Elasticities become increasingly negative with transaction size, reaching about -6.8 for the $700-and-above bin.
- Low-income households show larger percentage declines than high-income households.
- There is evidence of substitution to Newegg, but not to Best Buy or broad online-retail categories.
- The Amazon share of household retail spending falls after tax implementation.
- The results indicate that sales-tax noncollection was an economically important component of Amazon's price advantage.

8 Conclusion

8.1 Core conclusion

Baugh, Ben-David, and Park show that requiring Amazon to collect sales tax materially reduced household purchases from Amazon. The effect is economically large, persistent, and much stronger for large purchases. The paper therefore provides direct evidence that sales-tax collection can shape online retail demand and partly reallocate spending across retailers.

8.2 Economic intuition

Before tax collection, Amazon effectively offered a lower checkout price than retailers that collected sales tax. Once Amazon began collecting tax, its price advantage shrank. Consumers responded more strongly for large-ticket items because the dollar value of tax avoidance is larger. If switching or search costs are fixed, the incentive to comparison-shop becomes strongest when the purchase amount is high.

8.3 Incremental contribution revisited

The paper advances the online-commerce tax literature in three ways. First, it uses a real policy change rather than survey responses or within-platform seller comparisons. Second, it uses household-level transaction data that can track purchases before and after treatment. Third, it links the Amazon response to competitor outcomes, purchase-size elasticities, and household heterogeneity.

8.4 Implications for tax policy and competition

The findings imply that tax collection rules can materially affect the competitive position of online retailers. For states, requiring collection may raise tax revenue and reduce the implicit subsidy to out-of-state sellers. For retailers, the result shows that a tax advantage can meaningfully shift demand, especially for products with high prices and low comparison costs.

8.5 Limitations

The account-aggregator sample is not a random sample of the U.S. population. The data identify merchants but not products, so substitution to broad retailers is difficult to measure. State adoption of Amazon-tax laws is not randomized, even though the paper provides several checks against local economic confounds. Finally, the results apply to a period before later legal and institutional changes that expanded online sales-tax collection more broadly.

8.6 Takeaway

The enduring takeaway is that tax salience and tax enforcement at checkout matter for household behavior. Amazon's sales-tax advantage was not merely a legal technicality; it affected real purchasing decisions, especially for large purchases, and its removal changed the competitive landscape of online retail.